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CHAPTER 2 • ALGEBRA • TOPIC 12

Simultaneous Equations

Two equations, two unknowns. Match a coefficient and eliminate — or substitute the line into the curve. The calculator checks every answer.

**7**

VARIANTS

26

PAST-PAPER QUESTIONS

51

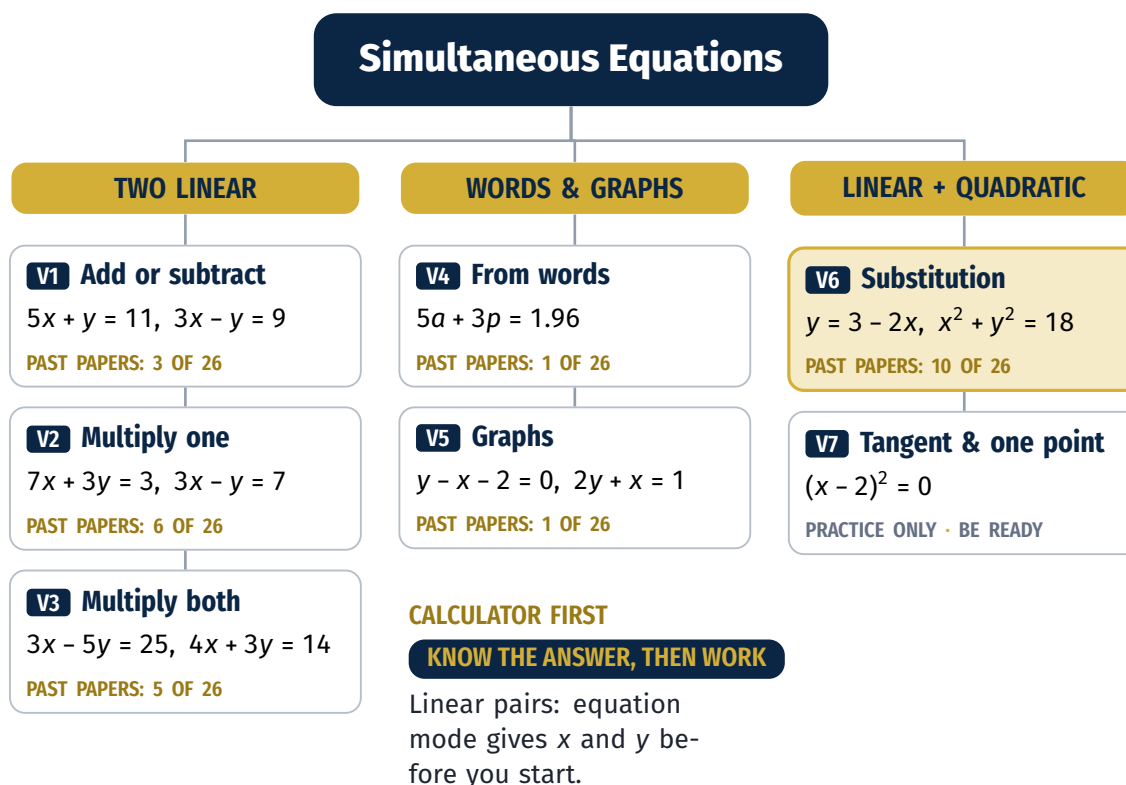
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Add or subtract **V2** Multiply one **V3** Multiply both **V4** From words**V5** Graphs **V6** Linear + quadratic **V7** Tangent & one point

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Simultaneous Equations at a glance



Your notes


★ How often does each variant come up?

26 past-paper questions (Jan 2020 → Nov 2025); 51 practice parts — grouped by type.

	PAST PAPERS (OUT OF 26)	PRACTICE PARTS
V1 Add or subtract	3	4
V2 Multiply one	6	6
V3 Multiply both	5	10
V4 From words	1	12
V5 Graphs	1	4
V6 Linear + quadratic	10	10
V7 Tangent & one point	0	5

WHAT THIS TELLS YOU Two questions in almost every series: a linear pair (14 parts, 3–4 marks) and a linear + quadratic (10 parts, 5 marks, near the end of the paper).

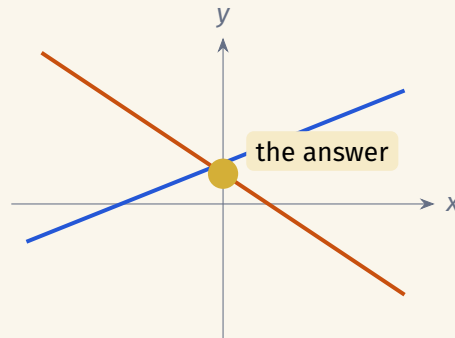
BEFORE YOU START

The big idea

Two equations, both true at once.

Each equation is a line. The answer is the point that is on **both** lines — where they **cross**.

With algebra: **remove one letter** (eliminate or substitute), solve for the other, then substitute back.



 Your notes

**Two ways to remove a letter****Elimination (linear pairs)**

$$5x + y = 11 \quad + \quad 3x - y = 9$$

Match one coefficient, then **add** (signs different) or **subtract** (signs the same).

Substitution (line + curve)

$$y = 3 - 2x \rightarrow x^2 + (3 - 2x)^2 = 18$$

Make one letter the subject of the **linear** equation, put it into the other.

**Dr Eslam's sign rule**

+5y and -5y
DIFFERENT SIGNS → ADD

3x and 3x
SAME SIGNS → SUBTRACT

**CALCULATOR FIRST, THEN CHECK**

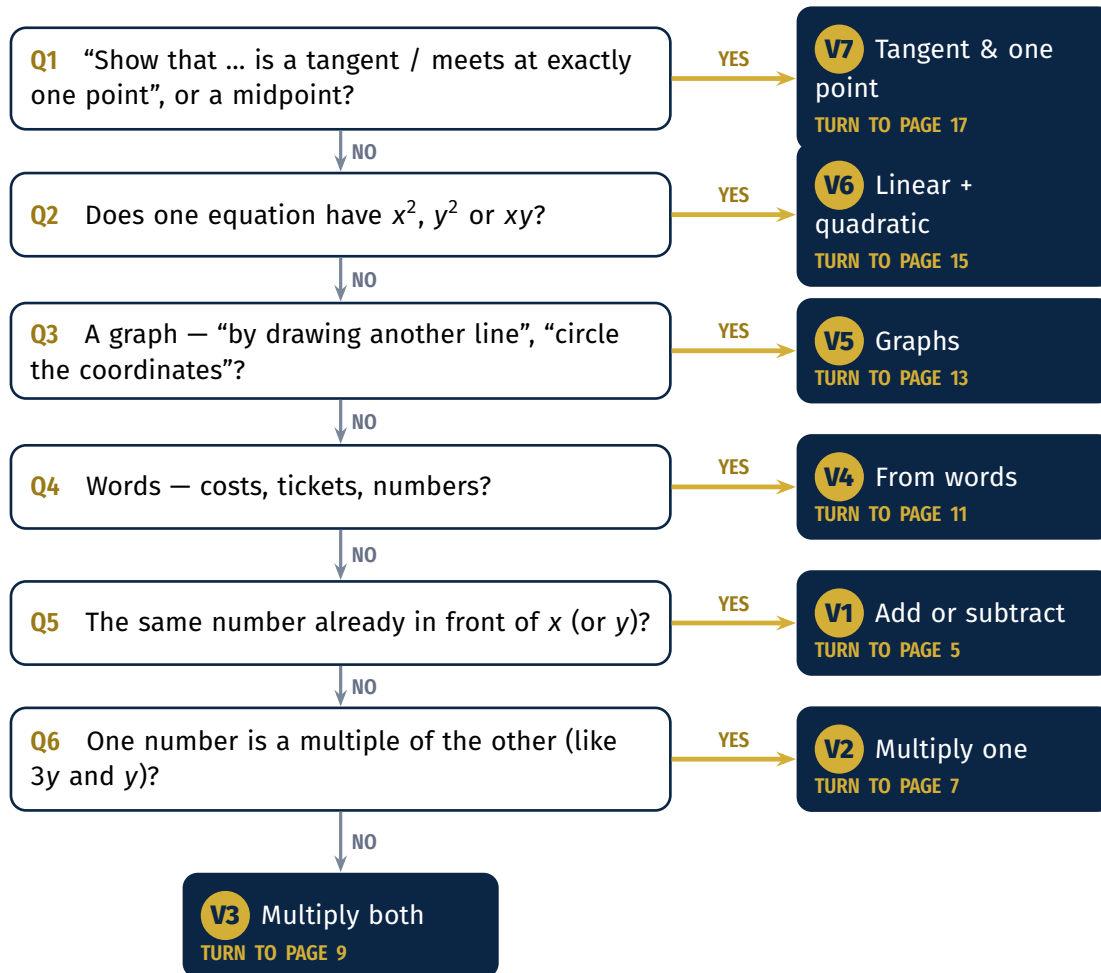
Calculator first: **MENU** → Equation → Simul Equation → **2** unknowns (fx-991ES: **MODE** **5** **1**). Enter the numbers of both equations — read x and y. Now you know the answer; write the algebra that gets there. “Show clear algebraic working” — the calculator alone scores 0.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Look at the two equations, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Add or Subtract

The same number is already in front of one letter: add or subtract to remove it.

SEEN IN PAST PAPERS

3 / 26

PRACTICE QUESTIONS: 4



Your notes

Came up in: Jan 2020 P2HR Q7 · Jun 2025 P2HR Q3 · Nov 2025 P2H Q2

? What is it?

In plain words

Look for a letter with the **same number** in front in both equations.

Different signs (+y, -y) → **add** the equations.

Same signs (3x, 3x) → **subtract** them.

That letter disappears; solve for the other, then substitute back.

IT LOOKS LIKE

$$5x + y = 11 \text{ \& } 3x - y = 9$$

$$3x + 2y = 10 \text{ \& } 3x - 4y = 16$$

$$\begin{array}{r}
 5x + y = 11 \\
 + \quad 3x - y = 9 \\
 \hline
 8x = 20
 \end{array}$$

+y AND -y CANCEL

How do I do it?

1 Calculator

Equation mode: know x and y first.

2 Add or subtract

Different signs → add. Same signs → subtract (bracket it).

3 Solve

One letter left: $8x = 20$, so $x = 2.5$.

4 Back-substitute

Put x into either equation to find y. Check in the other.

DIFFERENT → ADD · SAME → SUBTRACT Look at the signs of the matching terms.

V1 Add or Subtract

VARIANT 1 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO*+y in one, -y in the other*→ **Add the equations***3x in both*→ **Subtract:** $(3x + 2y) - (3x - 4y)$ *"Show clear algebraic working"*→ **Write the added/subtracted line****WORKED EXAMPLE****PAST PAPER**

November 2025 · Paper 2H · Q2 · 3 marks

Solve the simultaneous equations $5x + y = 11$ and $3x - y = 9$. Show clear algebraic working.

STEP 1 $x = 2.5, y = -1.5$

◀ calculator first

STEP 2 $8x = 20$

◀ different signs: add

STEP 3 $x = 2.5$

◀ solve

STEP 4 $5(2.5) + y = 11 \Rightarrow y = -1.5$

◀ back-substitute

ANSWER $x = 2.5, y = -1.5$ ✓ $3(2.5) - (-1.5) = 9$ ✓**WATCH OUT**✗ $(3x + 2y) - (3x - 4y) = -2y$ ✓ $(3x + 2y) - (3x - 4y) = 6y$

Subtracting $-4y$ means $+4y$. Put the second equation in a bracket.

**EXAM TIP**

Check your pair in the **other** equation – the one you did not use to find y .

**Check yourself**

calculator first · answers on page 19

1.1

Jan 2020 P2HR

Solve $3x + 5y = 6$,
 $7x - 5y = -11$

Answer:

1.2

Jun 2025 P2HR

Solve $3x + 2y = 10$,
 $3x - 4y = 16$

Answer:

1.3

Practice · Easy

Solve $2x + y = 18$,
 $x - y = 6$

Answer:

V2

VARIANT 2

Multiply One Equation

One coefficient fits into the other: multiply one equation to make them match.

SEEN IN PAST PAPERS

6 / 26

PRACTICE QUESTIONS: 6



Your notes

Came up in: May 2021 P2H Q8 • Jan 2022 P2H Q10 • Jun 2022 P2H Q10 • May 2023 P1H Q9 • Jun 2024 P2H Q10 • Jun 2025 P2H Q8

? What is it?

In plain words

Nothing matches yet — but one number goes into another: 3y and -y (×3), 3x and 6x (×2).

Multiply **every term** of that one equation (the right side too). Now it is V1.

IT LOOKS LIKE

$$7x + 3y = 3 \text{ \& } 3x - y = 7$$

$$6x + 4y = 1 \text{ \& } 3x + 5y = 8$$

$$\begin{array}{rcl} 3x - y = 7 & \xrightarrow{\times 3} & 9x - 3y = 21 \\ & + & 7x + 3y = 3 \\ \hline & & 16x = 24 \end{array}$$

Every term — the 7 becomes 21.

How do I do it?

1 Choose

Which letter is easier? -y and 3y: multiply by 3.

2 Multiply

Every term of that equation, including the number on the right.

3 Eliminate

Different signs → add; same → subtract. Solve.

4 Back-substitute

Find the other letter; check in both.

MULTIPLY EVERY TERM

The right-hand side too.

V2 Multiply One Equation

VARIANT 2 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO*-y against 3y*→ **×3 the first, then add***3x against 6x*→ **×2 the first, then subtract***Decimals: $3x + 5y = 3.1$* → **Same method – the calculator handles the decimals****WORKED EXAMPLE** **PAST PAPER**

June 2022 · Paper 2H · Q10 · 3 marks

Solve the simultaneous equations $7x + 3y = 3$ and $3x - y = 7$. Show clear algebraic working.

STEP 2 $9x - 3y = 21$

◀ **×3 the second equation**

STEP 3 $16x = 24 \Rightarrow x = 1.5$

◀ **add (different signs)**

STEP 4 $3(1.5) - y = 7 \Rightarrow y = -2.5$

◀ **back-substitute****ANSWER** $x = 1.5, y = -2.5$

✓ $7(1.5) + 3(-2.5) = 3$ ✓

**WATCH OUT**

✗ **×3: $9x - 3y = 7$**

✓ **×3: $9x - 3y = 21$**

The right-hand side must be multiplied too.

**EXAM TIP**

6 of the 26 past-paper questions.

Calculator first tells you $x = 1.5, y = -2.5$

— your algebra must land there.

**Check yourself**

calculator first · answers on page 19

2.1

Jun 2024 P2H

Solve $6x + 4y = 1$,
 $3x + 5y = 8$

Answer:

2.2

Jun 2025 P2H

Solve $3x + 5y = 8$,
 $4x + y = -3.5$

Answer:

2.3

May 2023 P1H

Solve $2x + 9y = 14.5$,
 $7x + 3y = 8$

Answer:

V3

VARIANT 3

Multiply Both Equations

Nothing fits: multiply both equations to reach a common number (the LCM).

SEEN IN PAST PAPERS

5 / 26

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Oct/Nov 2020 P1H Q12 • Oct/Nov 2020 P1HR Q12 • Jan 2021 P2HR Q12 • Jun 2022 P1HR Q14 • May 2024 P1HR Q12

? What is it?

In plain words

-5y and 3y: neither fits into the other. Use the **LCM** of 5 and 3, which is 15: multiply the first by 3 and the second by 5.

Now -15y and +15y — add.

IT LOOKS LIKE

$$3x - 5y = 25 \text{ \& } 4x + 3y = 14$$

$$2x + 7y = 17 \text{ \& } 5x + 3y = -1$$

$$\begin{array}{rcl}
 3x - 5y = 25 & \xrightarrow{\times 3} & 9x - 15y = 75 \\
 4x + 3y = 14 & \xrightarrow{\times 5} & 20x + 15y = 70 \\
 \hline
 & & 29x = 145
 \end{array}$$

How do I do it?

1 Pick a letter

Usually the one with opposite signs — then you add.

2 Find the LCM

5 and 3 → 15. Multiply each equation to get 15.

3 Eliminate

Add or subtract; solve for one letter.

4 Back-substitute

Use an original equation; check in the other.

LCM OF THE COEFFICIENTS Then it is V1.

V3 Multiply Both Equations

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO $-5y$ and $3y$ (no number fits)→ $\times 3$ and $\times 5: \pm 15y$ Decimals: $4x + 3y = 9.6$

→ Same method; keep the decimals exact

(4 marks)

→ M1 match, M1 eliminate, A1 one letter, A1 the other

WORKED EXAMPLE **PAST PAPER**

June 2022 · Paper 1HR · Q14 · 4 marks

Solve the simultaneous equations $3x - 5y = 25$ and $4x + 3y = 14$.
Show clear algebraic working.

STEP 2

$9x - 15y = 75, \quad 20x + 15y = 70$

◀ $\times 3$ and $\times 5$: LCM 15**STEP 3**

$29x = 145 \Rightarrow x = 5$

◀ add (different signs), solve

STEP 4

$3(5) - 5y = 25 \Rightarrow y = -2$

◀ back-substitute

ANSWER $x = 5, y = -2$ $\checkmark 4(5) + 3(-2) = 14 \checkmark$ **WATCH OUT**

$\times -5y = 10 \Rightarrow y = 2$

$\checkmark -5y = 10 \Rightarrow y = -2$

Positive $\div$ negative is negative. The calculator check catches it.

**EXAM TIP**

Choose the letter with **opposite** signs to eliminate — adding is safer than subtracting.

**Check yourself**

calculator first · answers on page 19

3.1

Jan 2021 P2HR

Solve $2x + 7y = 17$,
 $5x + 3y = -1$

Answer:

3.2

May 2024 P1HR

Solve $4x + 3y = 9.6$,
 $6x + 5y = 16.8$

Answer:

3.3

Oct/Nov 2020 P1H

Solve $7x - 2y = 34$,
 $3x + 5y = -3$

Answer:

V4

VARIANT 4

From Words

Prices, tickets, numbers: each sentence becomes one equation.

SEEN IN PAST PAPERS

1 / 26

PRACTICE QUESTIONS: 12



Your notes

Came up in: Nov 2021 P1H Q6

? What is it?**In plain words**

Choose a **letter for each unknown** (and say what it means). Each fact in the question gives **one equation**. Solve them (V1–V3), then **answer the question asked** — with units.

IT LOOKS LIKE

$5a + 3p = 1.96$

$2c + d = 5$

Alison: 5 apples, 3 pears
— \$1.96

$5a + 3p = 1.96$

Greg: 3 apples, 2 pears
— \$1.22

$3a + 2p = 1.22$

How do I do it?**1 Letters**

“Let a = cost of one apple, p = cost of one pear.”

2 Equations

One equation per fact. Units the same (all \$ or all cents).

3 Solve

Calculator first, then eliminate (V1–V3).

4 Answer the question

Often a final step: “10 apples and 10 pears cost ...”.

ONE FACT = ONE EQUATION Then finish what they asked.

V4 From Words

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"5 apples and 3 pears cost \$1.96" → $5a + 3p = 1.96$

"Work out how much ... pays" → **Solve, then answer that exact question**

"the sum is 40, the difference is 12" → $x + y = 40, x - y = 12$

WORKED EXAMPLE **PAST PAPER**

November 2021 · Paper 1H · Q6 · 5 marks

5 apples and 3 pears cost \$1.96. 3 apples and 2 pears cost \$1.22. How much do 10 apples and 10 pears cost?

STEP 2 $5a + 3p = 1.96, \quad 3a + 2p = 1.22$ ◀ one fact, one equation

STEP 3 $10a + 6p = 3.92, \quad 9a + 6p = 3.66 \Rightarrow a = 0.26$ ◀ $\times 2, \times 3$, subtract

STEP 3 $3(0.26) + 2p = 1.22 \Rightarrow p = 0.22$ ◀ back-substitute

STEP 4 $10(0.26) + 10(0.22) = 4.80$ ◀ answer the question

ANSWER \$4.80✓ $5(0.26) + 3(0.22) = 1.96$ ✓**WATCH OUT**

✗ stopping at $a = 0.26, p = 0.22$

✓ finishing: \$4.80

The question asks for 10 apples and 10 pears — the last mark is for that.

EXAM TIP

Write a key: " a = apple in \$". It stops you mixing up the letters at the end.

Check yourself

calculator first · answers on page 19

4.1

Dr Eslam

2 cakes and 1 drink cost £5; 1 cake and 3 drinks cost £10. Find each price.

Answer:

4.2

Dr Eslam

3 adult and 2 child tickets cost £34; 2 adult and 3 child cost £31. Find each price.

Answer:

4.3

Dr Eslam

Two numbers have sum 40 and difference 12. Find them.

Answer:

V5

VARIANT 5

Graphs: the Crossing Point

The solution is where the two lines cross.
Draw the missing line, read the point.

SEEN IN PAST PAPERS

1 / 26

PRACTICE QUESTIONS: 4


Your notes

Came up in: Nov 2023 P2H Q14

? What is it?

In plain words

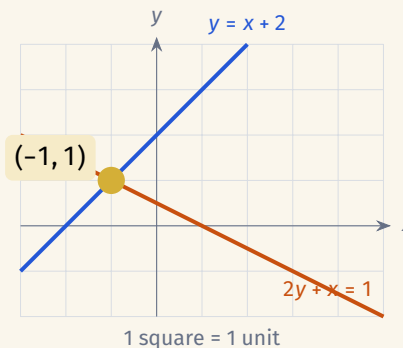
Each equation is a straight line.
The **crossing point** (x, y) is the solution.

“By drawing another line”:
rearrange to $y = \dots$, plot two or three points, draw it with a ruler, read the crossing.

IT LOOKS LIKE

$$y - x - 2 = 0 \text{ \& } 2y + x = 1$$

$$y = 3x \text{ \& } y = -1$$



How do I do it?

1 Rearrange

Write the new line as $y = \dots$: $y - x - 2 = 0 \rightarrow y = x + 2$.

2 Plot

Two or three points ($x = 0, 1, 2$). Ruler line across the grid.

3 Read

The crossing point: x and y (estimate if between lines).

4 Check

Put the point into both equations (or the calculator).

SOLUTION = CROSSING POINT Graphs give estimates; algebra gives exact.

V5 Graphs: the Crossing Point

VARIANT 5 · SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

“By drawing another straight line, solve ...”

→ Draw it, read the crossing

“Circle the coordinates of P” (two lines) → Solve the two equations

“find estimates of the solutions”

→ Read to the nearest small square

 WORKED EXAMPLE **PAST PAPER**

November 2023 · Paper 2H · Q14 · 3 marks

The graph of $2y + x = 1$ is drawn. By drawing another straight line, solve $y - x - 2 = 0$ and $2y + x = 1$.

STEP 1 $y = x + 2$

◀ rearrange the new line

STEP 2 $(0, 2), (1, 3), (-2, 0)$

◀ plot points, draw with a ruler

STEP 3 crossing at $(-1, 1)$

◀ read the point

STEP 4 $2(1) + (-1) = 1$ ✓

◀ check in the other equation

ANSWER $x = -1, y = 1$

✓ calculator: same

 WATCH OUT✗ plotting $y - x - 2 = 0$ as $y = x - 2$ ✓ $y = x + 2$ Move $-x - 2$ across: both signs flip. **EXAM TIP**

Draw the line across the **whole** grid — a short line may not reach the crossing.

 Check yourself

calculator first · answers on page 19

5.1 Practice · Easy

The lines $y = 3x$ and $y = -1$ meet at P. Find P.

Answer:

5.2 Practice · Easy

The lines $y = -4x$ and $y = 3$ meet at A. Find A.

Answer:

5.3 Dr Eslam

Where do $y = 2x - 1$ and $x + y = 5$ cross?

Answer:

V6

VARIANT 6 MOST COMMON IN THE EXAM

Linear + Quadratic: Substitution

Make one letter the subject of the line, substitute into the curve, solve the quadratic — two pairs.

SEEN IN PAST PAPERS

10 / 26

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Jan 2021 P1H Q19 · Jan 2021 P1HR Q17 · May 2021 P2H Q19 · Jan 2022 P1H Q19 · Jun 2022 P2HR Q21 · Jun 2023 P2H Q21 · Jun 2024 P2HR Q22 · Nov 2024 P2H Q22 · Jun 2025 P1HR Q22 · Jun 2025 P1H Q22

? What is it?

In plain words

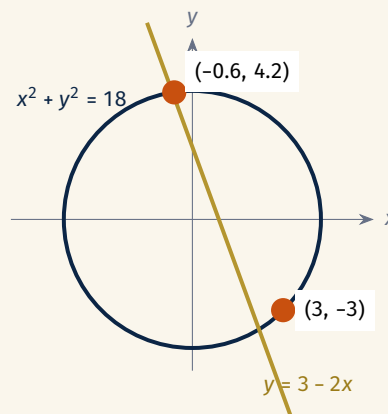
One equation is a **line**, the other has x^2 or y^2 (a circle or curve). A line usually cuts a curve **twice** — so there are **two pairs** of answers.

Substitute the line into the curve. You get a quadratic: ERCS, solve, then find the other letter for each.

IT LOOKS LIKE

$$y = 3 - 2x \text{ \& } x^2 + y^2 = 18$$

$$x = 3y - 1 \text{ \& } 2x^2 + 3y^2 = 11$$



How do I do it?

1 Subject from the line

Use the letter with coefficient 1: $x - 6y = 5 \rightarrow x = 6y + 5$.

2 Substitute

Put it into the curve — in a **bracket**: $x^2 + (3 - 2x)^2 = 18$.

3 ERCS + solve

Expand, rearrange to $= 0$, factorise or the formula.

4 Pair them up

Each value into the **line**: two pairs (x, y) .

LINE INTO CURVE, IN A BRACKET Two answers $\rightarrow$ two pairs.

V6 Linear + Quadratic: Substitution

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DOOne equation with x^2 , y^2 or xy → **Substitute the line into it** $x - 6y = 5$ (coefficient 1)→ $x = 6y + 5$ — **easiest to substitute**

“Show clear algebraic working” (5 marks)

→ **Substitution line, quadratic = 0, both pairs****WORKED EXAMPLE** **PAST PAPER**

May 2021 · Paper 2H · Q19 · 5 marks

Solve the simultaneous equations $y = 3 - 2x$ and $x^2 + y^2 = 18$. Show clear algebraic working.**STEP 2** $x^2 + (3 - 2x)^2 = 18$ ◀ substitute, in a bracket**STEP 3** $x^2 + 9 - 12x + 4x^2 = 18 \Rightarrow 5x^2 - 12x - 9 = 0$ ◀ expand, rearrange**STEP 3** $(5x + 3)(x - 3) = 0 \Rightarrow x = 3$ or $x = -0.6$ ◀ factorise (calculator roots)**STEP 4** $y = 3 - 2(3) = -3$, $y = 3 - 2(-0.6) = 4.2$ ◀ each x into the line**ANSWER** $(3, -3)$ and $(-0.6, 4.2)$ $\checkmark 9 + 9 = 18 \checkmark$, $0.36 + 17.64 = 18 \checkmark$ **WATCH OUT**

$\times (3 - 2x)^2 = 9 + 4x^2$

$\checkmark (3 - 2x)^2 = 9 - 12x + 4x^2$

A squared bracket has a middle term — expand it fully.

EXAM TIP10 of the 26 past-paper questions, 5 marks each. Find y from the **line**, not the curve — the curve can give extra wrong values.**Check yourself**

calculator first · answers on page 19

6.1 Jun 2025 P1HRSolve $x^2 + y^2 = 41$,
 $2x + y = 3$

Answer:

6.2 Nov 2024 P2HSolve $x^2 + y^2 + y = 3$,
 $x + 2 = y$

Answer:

6.3 Jun 2023 P2HSolve $2x^2 + 3y^2 = 11$,
 $x = 3y - 1$

Answer:

V7

VARIANT 7

Tangent & One Point

Show a line touches a curve once:
substitute, the quadratic is a perfect square.

SEEN IN PAST PAPERS

0 / 26

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 5


Your notes
? What is it?**In plain words**

Substitute as in V6. The quadratic tells you how many times they meet:

- two roots → **crosses twice**
- one repeated root, $(x - 2)^2 = 0 \rightarrow$ **touches once: a tangent**
- no roots → misses.

Midpoint of the two crossings:
average the coordinates.

IT LOOKS LIKE

$$y = 2x + 1 \text{ \& } y = x^2 - 2x + 5$$

**How do I do it?****1 Substitute**

Line into the curve (V6).

2 Quadratic = 0ERCS: e.g. $x^2 - 4x + 4 = 0$.**3 Perfect square**Factorise: $(x - 2)^2 = 0$ — one repeated root (or $b^2 - 4ac = 0$).**4 Conclude**

“One point of contact, so the line is a tangent”, with the point.

ONE REPEATED ROOT = TANGENT

Say it in a sentence.

V7 Tangent & One Point

VARIANT 7 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Show that the line is a tangent to the circle”

→ **Substitute; show $(\dots)^2 = 0$**

“meets at exactly one point”

→ **The same — one repeated root**

“the midpoint of PQ”

→ **Find P and Q, average the coordinates**

WORKED EXAMPLE DR ESAM

Algebraic method · 4 marks

Show that the line $y = 2x + 1$ meets the curve $y = x^2 - 2x + 5$ at exactly one point, and find it.

STEP 1 $x^2 - 2x + 5 = 2x + 1$ ◀ substitute (both equal y)

STEP 2 $x^2 - 4x + 4 = 0$ ◀ rearrange to $= 0$

STEP 3 $(x - 2)^2 = 0 \Rightarrow x = 2$ (repeated) ◀ a perfect square: one root

STEP 4 $y = 2(2) + 1 = 5$ ◀ the point

ANSWER one repeated root — they meet once, at (2, 5) ✓ $4 - 4 + 5 = 5$ ✓

WATCH OUT

× “ $x = 2$, so it is a tangent”

✓ “ $(x - 2)^2 = 0$: a repeated root, so one point”

The reason is the **repeated** root — write it.

EXAM TIP

If it does not factorise neatly, show $b^2 - 4ac = 0$ instead: $16 - 16 = 0$.

Check yourself

calculator first · answers on page 19

7.1 Practice · Very hard

Show that $x - 2y = 10$ is a tangent to $x^2 + y^2 = 20$.

Answer:

7.2 Dr Esam

$y = x - 3$ meets $x^2 + y^2 = 5$ at P and Q. Find the midpoint of PQ.

Answer:

7.3 Dr Esam

Show that $y = 4x - 4$ is a tangent to $y = x^2$.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $x = -0.5, y = 1.5$

1.2 $x = 4, y = -1$

1.3 $x = 8, y = 2$

2.1 $x = -1.5, y = 2.5$

2.2 $x = -1.5, y = 2.5$

2.3 $x = 0.5, y = 1.5$

3.1 $x = -2, y = 3$

3.2 $x = -1.2, y = 4.8$

3.3 $x = 4, y = -3$

4.1 cake £1, drink £3

4.2 adult £8, child £5

4.3 26 and 14

5.1 $\left(-\frac{1}{3}, -1\right)$

5.2 $\left(-\frac{3}{4}, 3\right)$

5.3 $(2, 3)$

6.1 $(4, -5)$ and $(-1.6, 6.2)$

6.2 $(-1, 1)$ and $(-1.5, 0.5)$

6.3 $(2, 1)$ and $\left(-\frac{16}{7}, -\frac{3}{7}\right)$

7.1 $(y + 4)^2 = 0$: one point $(2, -4)$

7.2 $(1.5, -1.5)$

7.3 $(x - 2)^2 = 0$: one point $(2, 4)$

THE WHOLE TOPIC ON ONE PAGE

Simultaneous Equations cheat sheet


 Your notes
V1 Add or subtract 3/26

$$5x + y = 11, 3x - y = 9 \Rightarrow 8x = 20$$

Different signs add; same signs subtract.

V2 Multiply one 6/26

$$3x - y = 7 \quad (\times 3)$$

Multiply every term, then eliminate.

V3 Multiply both 5/26

$$\times 3, \times 5 : \pm 15y$$

LCM of the coefficients.

V4 From words 1/26

$$5a + 3p = 1.96$$

One fact, one equation; answer the question.

V5 Graphs 1/26crossing at $(-1, 1)$

Draw the missing line; read the crossing.

V6 Linear + quadratic 10/26

$$x^2 + (3 - 2x)^2 = 18$$

Line into curve; two pairs; y from the line.

V7 Tangent

$$(x - 2)^2 = 0$$

One repeated root = one point.

**TOP 3 LOST MARKS**

1. $-(-4y)$ sign slip when subtracting $\times$
2. not multiplying the right side $\times$
3. $(3 - 2x)^2 = 9 + 4x^2 \times$

BOTTOM LINE Calculator first $\rightarrow$ match & eliminate (linear) or substitute (line + curve) $\rightarrow$ back-substitute $\rightarrow$ check in both.